

Example usage of Mega2R

Robert V. Baron and Daniel E. Weeks

June 13, 2017, 11:08

Contents

1	Load Libraries	1
2	Input directory and files	1
3	Analysis	1
3.1	Using the mega2 R package	1
3.2	Using the mega2pedgene R package	3
3.3	Using the mega2vcf R package	8
3.4	Using the mega2genabel R package	9
4	Regression	10
4.1	GenABEL regression	10
5	Session Information	11

1 Load Libraries

2 Input directory and files

```
# Print the working directory
getwd()
```

```
## [1] "/Users/dweeks/Temporary/test_mega2r"
```

The input files used in this analysis are ... and can be found in ...

3 Analysis

3.1 Using the mega2 R package

Create the Mega2 database “seqsimr.db” using mega2

```
mega2 Mega2.BATCH.seqsimr
```

```
library(mega2)
# Read the Mega2 database into R
ENV = read.Mega2DB("seqsimr.db", verbose = TRUE)
```

```
## int_table      34  3
## int_table      pId key value
##
## pedigree_table  20  8
## pedigree_table pId Num EntryCnt   Name   PedPre OriginalID origped pedigree_link
```

```

##
## person_table 1380 11
## person_table pId UniqueID OrigID FamName PerPre ID Father Mother Sex pedigree_link person_
##
## pedigree_brkloop_table 20 8
## pedigree_brkloop_table pId Num EntryCnt Name PedPre OriginalID origped pedigree_link
##
## person_brkloop_table 1380 11
## person_brkloop_table pId UniqueID OrigID FamName PerPre ID Father Mother Sex pedigree_link
##
## locus_table 1001 5
## locus_table pId LocusName Type AlleleCnt locus_link
##
## allele_table 2002 5
## allele_table pId AlleleName Frequency indexX locus_link
##
## marker_table 1000 7
## marker_table pId MarkerName pos_avg pos_female pos_male chromosome locus_link
##
## markerscheme_table 1000 4
## markerscheme_table pId key allele1 allele2
##
## map_table 2000 6
## map_table pId marker map position pos_female pos_male
##
## mapnames_table 2 6
## mapnames_table pId map sex_averaged_map male_sex_map female_sex_map name
##
## traitaff_table 1 4
## traitaff_table pId ClassCnt PenCnt locus_link
##
## affectclass_table 1 9
## affectclass_table pId MaleDef FemaleDef AutoDef MalePen FemalePen AutoPen locus_link class_1
##
## phenotype_table 1380 4
## phenotype_table pId person_link bytes data
##
## genotype_table 1380 5
## genotype_table pId person_link chr bytes data

```

Examine the tables that have been created

```
ls(ENV)
```

```

## [1] "affectclass_table" "allele_table"
## [3] "entrezGene"       "fam"
## [5] "int_table"        "locus_allele_table"
## [7] "locus_table"      "LocusCnt"
## [9] "map_table"        "mapnames_table"
## [11] "MARKER_SCHEME"    "marker_table"
## [13] "markers"          "markerscheme_table"
## [15] "pedigree_brkloop_table" "pedigree_table"
## [17] "person_brkloop_table" "person_table"
## [19] "PhenoCnt"         "phenotype_table"
## [21] "refIndices"       "refRanges"
## [23] "traitaff_table"   "txdb"

```

```
## [25] "unified_genotype_table" "verbose"
# A more informative overview
showMega2ENV()

## locus count: 1001; phenotype count: 1; compression: 2 bits
## marker count: 1000; sample count: 1380
##
## genetic and physical maps:
## map name map number
## 1 Map 0
## 2 BP 1
##
## Phenotypes:
## Index Name Type
## 1 1 default affection
##
## basic tables:
## rows cols
## affectclass_table 1 9
## allele_table 2002 5
## int_table 34 3
## locus_table 1001 5
## map_table 2000 6
## mapnames_table 2 6
## marker_table 1000 8
## markerscheme_table 1000 4
## pedigree_brkloop_table 20 8
## pedigree_table 20 8
## person_brkloop_table 1380 11
## person_table 1380 11
## phenotype_table 1380 4
## traitaff_table 1 4
##
## derived tables:
## rows cols
## fam 1380 8
## markers 1000 5
## unified_genotype_table 1380 2

# Use standard R commands to examine the created data frames
str(ENV$locus_table)

## 'data.frame': 1001 obs. of 5 variables:
## $ pId : int 1 2 3 4 5 6 7 8 9 10 ...
## $ LocusName : chr "default" "snp1" "snp2" "snp3" ...
## $ Type : int 2 4 4 4 4 4 4 4 4 4 ...
## $ AlleleCnt : int 2 2 2 2 2 2 2 2 2 2 ...
## $ locus_link: int 0 1 2 3 4 5 6 7 8 9 ...
```

3.2 Using the mega2pedgene R package

```

library(mega2pedgene)
ENV = init_pedgene("seqsimr.db", verbose = TRUE)

## int_table      34  3
## int_table      pId key value
##
## pedigree_table 20  8
## pedigree_table pId Num EntryCnt      Name      PedPre  OriginalID  origped pedigree_link
##
## person_table 1380    11
## person_table pId UniqueID      OrigID  FamName PerPre  ID  Father  Mother  Sex pedigree_link  person_
##
## pedigree_brkloop_table 20  8
## pedigree_brkloop_table pId Num EntryCnt      Name      PedPre  OriginalID  origped pedigree_link
##
## person_brkloop_table 1380    11
## person_brkloop_table pId UniqueID      OrigID  FamName PerPre  ID  Father  Mother  Sex pedigree_link
##
## locus_table 1001    5
## locus_table pId LocusName      Type      AlleleCnt  locus_link
##
## allele_table 2002    5
## allele_table pId AlleleName      Frequency      indexX  locus_link
##
## marker_table 1000    7
## marker_table pId MarkerName      pos_avg pos_female pos_male      chromosome locus_link
##
## markerscheme_table 1000    4
## markerscheme_table pId key allele1 allele2
##
## map_table 2000    6
## map_table pId marker map position      pos_female pos_male
##
## mapnames_table 2    6
## mapnames_table pId map sex_averaged_map      male_sex_map      female_sex_map  name
##
## traitaff_table 1    4
## traitaff_table pId ClassCnt      PenCnt  locus_link
##
## affectclass_table 1    9
## affectclass_table pId MaleDef FemaleDef      AutoDef MalePen FemalePen      AutoPen locus_link  class_l
##
## phenotype_table 1380    4
## phenotype_table pId person_link bytes      data
##
## genotype_table 1380    5
## genotype_table pId person_link chr bytes      data

# applyFnToRanges(DOpedgene, ENV$refRanges, ENV$refIndices, envir = ENV)
# applyFnToGenes(DOpedgene, genes_arg = c('FAM5C', 'TRABD2B'), envir = ENV)
applyFnToGenes(DOpedgene, genes_arg = c("GDAP2"), envir = ENV)

##

## 'select()' returned 1:1 mapping between keys and columns

```

'select()' returned 1:many mapping between keys and columns

```
## GDAP2 snp147763 1345 35 0
## GDAP2 snp147766 1192 184 4
## GDAP2 snp147770 1192 184 4
## GDAP2 snp147771 1172 196 12
## GDAP2 snp147778 393 714 273
## GDAP2 snp147785 1251 128 1
## GDAP2 snp147786 671 595 114
## GDAP2 snp147787 1378 2 0
## GDAP2 snp147789 787 518 75
## GDAP2 snp147794 1168 204 8
## GDAP2 snp147798 1312 68 0
## GDAP2 snp147807 1266 111 3
## GDAP2 snp147810 1376 4 0
## GDAP2 snp147817 810 506 64
## GDAP2 snp147818 120 614 646
## GDAP2 snp147823 1372 8 0
## GDAP2 snp147825 405 718 257
## GDAP2 snp147826 1354 26 0
## GDAP2 snp147829 1266 111 3
## GDAP2 snp147835 1168 204 8
## GDAP2 snp147837 810 506 64
## GDAP2 snp147840 1168 204 8
## GDAP2 snp147841 892 421 67
## GDAP2 snp147845 1376 4 0
## GDAP2 snp147850 810 506 64
## GDAP2 snp147853 1316 64 0
## GDAP2 snp147855 1370 10 0
## GDAP2 snp147856 1372 8 0
## GDAP2 snp147859 1374 6 0
## GDAP2 snp147861 406 718 256
## GDAP2 snp147866 64 506 810
## GDAP2 snp147868 1355 25 0
## GDAP2 snp147871 652 608 120
## GDAP2 snp147875 1378 2 0
## GDAP2 snp147886 1329 51 0
## GDAP2 snp147889 1370 10 0
## GDAP2 snp147892 1366 14 0
## GDAP2 snp147896 646 614 120
## GDAP2 snp147899 1351 29 0
## GDAP2 snp147906 120 614 646
## GDAP2 snp147907 1266 111 3
## GDAP2 snp147909 1357 23 0
## GDAP2 snp147910 120 614 646
## GDAP2 snp147912 1183 189 8
## GDAP2 snp147917 1246 129 5
## GDAP2 snp147919 1371 9 0
## GDAP2 snp147920 1240 137 3
## GDAP2 snp147923 47 468 865
## GDAP2 snp147928 1354 26 0
## GDAP2 snp147933 1345 35 0
## GDAP2 snp147935 1378 2 0
## GDAP2 snp147936 654 609 117
```

```

## GDAP2 snp147938 810 506 64
## GDAP2 snp147940 1364 16 0
## GDAP2 snp147941 1315 65 0
## GDAP2 snp147942 1375 5 0
## GDAP2 snp147943 1368 12 0
## GDAP2 snp147945 1378 2 0
## GDAP2 snp147946 1179 193 8
## GDAP2 snp147949 117 609 654
## GDAP2 snp147953 1240 137 3
## GDAP2 snp147955 1377 3 0
## GDAP2 snp147956 1313 67 0
## GDAP2 snp147957 810 506 64
## GDAP2 snp147958 68 424 888
## GDAP2 snp147959 1354 26 0
## GDAP2 snp147961 559 654 167
## GDAP2 snp147962 1331 49 0
## GDAP2 snp147963 1179 193 8
## GDAP2 snp147964 1376 4 0
## GDAP2 snp147967 117 609 654
## GDAP2 snp147973 120 614 646
## GDAP2 snp147976 1359 21 0
## GDAP2 snp147978 1368 12 0
## GDAP2 snp147980 1378 2 0
## GDAP2 snp147982 1370 10 0
## GDAP2 snp147983 1340 40 0
## GDAP2 snp147984 1378 2 0
## GDAP2 snp147989 1238 139 3
## GDAP2 snp147992 643 623 114
## GDAP2 snp148002 1168 204 8
## GDAP2 snp148004 1330 50 0
## GDAP2 snp148005 669 606 105
## GDAP2 snp148008 1251 128 1
## GDAP2 snp148011 111 602 667
## GDAP2 snp148012 1152 221 7
## GDAP2 snp148017 1369 11 0
## GDAP2 snp148020 1357 23 0
## GDAP2 snp148022 111 602 667
## GDAP2 snp148024 111 602 667
## GDAP2 snp148025 114 607 659
## GDAP2 snp148026 114 607 659
## GDAP2 snp148032 1377 3 0
## GDAP2 snp148034 1184 189 7
## GDAP2 snp148036 1238 139 3
## GDAP2 snp148037 1353 27 0
## GDAP2 snp148040 111 602 667
## GDAP2 snp148041 1235 141 4
## GDAP2 snp148042 413 714 253
## GDAP2 snp148043 111 602 667
## 1 GDAP2 100 118406107 118472302 0.732482 0.8535341 0.2916285 0.3511011 0.1993698 0.2153674 1
## GDAP2 snp147763 1345 35 0
## GDAP2 snp147766 1192 184 4
## GDAP2 snp147770 1192 184 4
## GDAP2 snp147771 1172 196 12
## GDAP2 snp147778 393 714 273

```

```

## GDAP2 snp147785 1251 128 1
## GDAP2 snp147786 671 595 114
## GDAP2 snp147787 1378 2 0
## GDAP2 snp147789 787 518 75
## GDAP2 snp147794 1168 204 8
## GDAP2 snp147798 1312 68 0
## GDAP2 snp147807 1266 111 3
## GDAP2 snp147810 1376 4 0
## GDAP2 snp147817 810 506 64
## GDAP2 snp147818 120 614 646
## GDAP2 snp147823 1372 8 0
## GDAP2 snp147825 405 718 257
## GDAP2 snp147826 1354 26 0
## GDAP2 snp147829 1266 111 3
## GDAP2 snp147835 1168 204 8
## GDAP2 snp147837 810 506 64
## GDAP2 snp147840 1168 204 8
## GDAP2 snp147841 892 421 67
## GDAP2 snp147845 1376 4 0
## GDAP2 snp147850 810 506 64
## GDAP2 snp147853 1316 64 0
## GDAP2 snp147855 1370 10 0
## GDAP2 snp147856 1372 8 0
## GDAP2 snp147859 1374 6 0
## GDAP2 snp147861 406 718 256
## GDAP2 snp147866 64 506 810
## GDAP2 snp147868 1355 25 0
## GDAP2 snp147871 652 608 120
## GDAP2 snp147875 1378 2 0
## GDAP2 snp147886 1329 51 0
## GDAP2 snp147889 1370 10 0
## GDAP2 snp147892 1366 14 0
## GDAP2 snp147896 646 614 120
## GDAP2 snp147899 1351 29 0
## GDAP2 snp147906 120 614 646
## GDAP2 snp147907 1266 111 3
## GDAP2 snp147909 1357 23 0
## GDAP2 snp147910 120 614 646
## GDAP2 snp147912 1183 189 8
## GDAP2 snp147917 1246 129 5
## GDAP2 snp147919 1371 9 0
## GDAP2 snp147920 1240 137 3
## GDAP2 snp147923 47 468 865
## GDAP2 snp147928 1354 26 0
## GDAP2 snp147933 1345 35 0
## GDAP2 snp147935 1378 2 0
## GDAP2 snp147936 654 609 117
## GDAP2 snp147938 810 506 64
## GDAP2 snp147940 1364 16 0
## GDAP2 snp147941 1315 65 0
## GDAP2 snp147942 1375 5 0
## GDAP2 snp147943 1368 12 0
## GDAP2 snp147945 1378 2 0
## GDAP2 snp147946 1179 193 8

```

```
## GDAP2 snp147949 117 609 654
## GDAP2 snp147953 1240 137 3
## GDAP2 snp147955 1377 3 0
## GDAP2 snp147956 1313 67 0
## GDAP2 snp147957 810 506 64
## GDAP2 snp147958 68 424 888
## GDAP2 snp147959 1354 26 0
## GDAP2 snp147961 559 654 167
## GDAP2 snp147962 1331 49 0
## GDAP2 snp147963 1179 193 8
## GDAP2 snp147964 1376 4 0
## GDAP2 snp147967 117 609 654
## GDAP2 snp147973 120 614 646
## GDAP2 snp147976 1359 21 0
## GDAP2 snp147978 1368 12 0
## GDAP2 snp147980 1378 2 0
## GDAP2 snp147982 1370 10 0
## GDAP2 snp147983 1340 40 0
## GDAP2 snp147984 1378 2 0
## GDAP2 snp147989 1238 139 3
## GDAP2 snp147992 643 623 114
## GDAP2 snp148002 1168 204 8
## GDAP2 snp148004 1330 50 0
## GDAP2 snp148005 669 606 105
## GDAP2 snp148008 1251 128 1
## GDAP2 snp148011 111 602 667
## GDAP2 snp148012 1152 221 7
## GDAP2 snp148017 1369 11 0
## GDAP2 snp148020 1357 23 0
## GDAP2 snp148022 111 602 667
## GDAP2 snp148024 111 602 667
## GDAP2 snp148025 114 607 659
## GDAP2 snp148026 114 607 659
## GDAP2 snp148032 1377 3 0
## GDAP2 snp148034 1184 189 7
## GDAP2 snp148036 1238 139 3
## GDAP2 snp148037 1353 27 0
## GDAP2 snp148040 111 602 667
## GDAP2 snp148041 1235 141 4
## GDAP2 snp148042 413 714 253
## GDAP2 snp148043 111 602 667
## 1 GDAP2 100 118419844 118472302 0.732482 0.8535341 0.2916285 0.3511011 0.1993698 0.2153674 1
```

```
system("cat k_Schaid_rare.txt", intern = TRUE)
```

```
## [1] "1 GDAP2 100 118406107 118472302 0.73248204056759 0.853534064929714 0.291628514463057 0.351101149
## [2] "1 GDAP2 100 118419844 118472302 0.73248204056759 0.853534064929714 0.291628514463057 0.351101149
```

3.3 Using the mega2vcf R package

```
dir.create("vcfr")
```

```
## Warning in dir.create("vcfr"): 'vcfr' already exists
```



```
library(mega2vcf)
Mega2VCF("vcfr/vcf")

## .

system("ls vcfr", intern = TRUE)

## [1] "vcf.fam" "vcf.freq" "vcf.map" "vcf.pen" "vcf.phe" "vcf.vcf"
```

3.4 Using the mega2genabel R package

```
library(mega2genabel)

## Loading required package: GenABEL
## Loading required package: MASS
## Loading required package: GenABEL.data
seqsingwaa = Mega2GenABEL("MEGA")

## .

## Reading individual ids from file 'MEGA.tfam' ...
## ... done. Read 1380 individual ids from file 'MEGA.tfam'
## Reading genotypes from file 'MEGA.tped' ...
## ...done. Read 1000 SNPs from file 'MEGA.tped'
## Writing to file 'MEGAtped.raw' ...
## ... done.
## ids loaded...
## marker names loaded...
## chromosome data loaded...
## map data loaded...
## allele coding data loaded...
## strand data loaded...
## genotype data loaded...
## snp.data object created...
## assignment of gwaa.data object FORCED; X-errors were not checked!

system("ls MEGA.*", intern = TRUE)

## [1] "MEGA.phe" "MEGA.tfam" "MEGA.tped"

str(seqsingwaa)

## Formal class 'gwaa.data' [package "GenABEL"] with 2 slots
## ..@ phdata:'data.frame': 1380 obs. of 3 variables:
## .. ..$ id : chr [1:1380] "1SAP039_H05-0107" "1SAP039_H05-0106" "1SAP039_SAP039F14" "1SAP039_SAP039F15" ...
## .. ..$ sex : int [1:1380] 0 1 1 1 1 0 0 0 0 0 ...
## .. ..$ default: int [1:1380] 1 1 1 1 1 1 1 1 1 1 ...
## ..@ gtdata:Formal class 'snp.data' [package "GenABEL"] with 11 slots
## .. ..@ nbytes : num 345
## .. ..@ nids : int 1380
## .. ..@ nsnps : int 1000
## .. ..@ idnames : chr [1:1380] "1SAP039_H05-0107" "1SAP039_H05-0106" "1SAP039_SAP039F14" "1SAP039_SAP039F15" ...
## .. ..@ snpnames : chr [1:1000] "snp1" "snp2" "snp3" "snp4" ...
## .. ..@ chromosome: Factor w/ 1 level "1": 1 1 1 1 1 1 1 1 1 1 ...
```

```
## .. .. .- attr(*, "names")= chr [1:1000] "snp1" "snp2" "snp3" "snp4" ...
## .. .. .@ map : Named num [1:1000] 2 3 4 5 5 6 8 11 12 17 ...
## .. .. .- attr(*, "names")= chr [1:1000] "snp1" "snp2" "snp3" "snp4" ...
## .. .. .@ coding :Formal class 'snp.coding' [package "GenABEL"] with 1 slot
## .. .. . .Data: raw [1:1000] 01 01 01 01 ...
## .. .. .@ strand :Formal class 'snp.strand' [package "GenABEL"] with 1 slot
## .. .. . .Data: raw [1:1000] 00 00 00 00 ...
## .. .. .@ male : Named int [1:1380] 0 1 1 1 1 0 0 0 0 0 ...
## .. .. .- attr(*, "names")= chr [1:1380] "1SAP039_H05-0107" "1SAP039_H05-0106" "1SAP039_SAP039F
## .. .. .@ gtps :Formal class 'snp.mx' [package "GenABEL"] with 1 slot
## .. .. . .Data: raw [1:345, 1:1000] 59 55 55 55 ...
```

4 Regression

4.1 GenABEL regression

```
library(GenABEL)
data(srdta)
dmpPed()
system("mega2 MEGA2.BATCH.srdta")
ENV = read.Mega2DB("srdta.db")
mega = Mega2GenABEL("srdta2")
```

```
## Reading individual ids from file 'srdta2.tfam' ...
## ... done. Read 2500 individual ids from file 'srdta2.tfam'
## Reading genotypes from file 'srdta2.tped' ...
## ...done. Read 833 SNPs from file 'srdta2.tped'
## Writing to file 'srdta2tped.raw' ...
## ... done.
## ids loaded...
## marker names loaded...
## chromosome data loaded...
## map data loaded...
## allele coding data loaded...
## strand data loaded...
## genotype data loaded...
## snp.data object created...
## assignment of gwaa.data object FORCED; X-errors were not checked!
```

```
Mega2GenABELtst()
```

```
## [1] TRUE
## [1] TRUE
## [1] TRUE
## [1] TRUE
## [1] TRUE
## [1] TRUE
## [1] TRUE
## [1] TRUE
## [1] TRUE
## [1] TRUE
## [1] TRUE
```

```
## [1] TRUE
## [1] TRUE
## [1] TRUE
```

5 Session Information

```
sessionInfo()
```

```
## R version 3.3.2 (2016-10-31)
## Platform: x86_64-apple-darwin13.4.0 (64-bit)
## Running under: OS X Mavericks 10.9.5
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] mega2genabel_0.5.0 GenABEL_1.8-0      GenABEL.data_1.0.0
## [4] MASS_7.3-45       mega2vcf_0.5.0     mega2pedgene_0.5.0
## [7] mega2_0.5.0       knitr_1.15.1
##
## loaded via a namespace (and not attached):
## [1] Rcpp_0.12.9
## [2] formatR_1.4
## [3] GenomeInfoDb_1.10.3
## [4] XVector_0.14.0
## [5] GenomicFeatures_1.26.3
## [6] bitops_1.0-6
## [7] tools_3.3.2
## [8] zlibbioc_1.20.0
## [9] biomaRt_2.30.0
## [10] digest_0.6.12
## [11] RSQLite_1.1-2
## [12] evaluate_0.10
## [13] memoise_1.0.0
## [14] lattice_0.20-34
## [15] TxDb.Hsapiens.UCSC.hg19.knownGene_3.2.2
## [16] Matrix_1.2-8
## [17] DBI_0.5-1
## [18] yaml_2.1.14
## [19] parallel_3.3.2
## [20] CompQuadForm_1.4.3
## [21] rtracklayer_1.34.2
## [22] kinship2_1.6.4
## [23] stringr_1.2.0
## [24] Biostrings_2.42.1
## [25] S4Vectors_0.12.1
## [26] IRanges_2.8.1
## [27] stats4_3.3.2
## [28] rprojroot_1.2
```

```
## [29] grid_3.3.2
## [30] Biobase_2.34.0
## [31] pedgene_2.9
## [32] AnnotationDbi_1.36.2
## [33] BiocParallel_1.8.1
## [34] XML_3.98-1.5
## [35] survival_2.40-1
## [36] rmarkdown_1.3
## [37] org.Hs.eg.db_3.4.0
## [38] magrittr_1.5
## [39] GenomicAlignments_1.10.0
## [40] Rsamtools_1.26.1
## [41] backports_1.0.5
## [42] htmltools_0.3.5
## [43] BiocGenerics_0.20.0
## [44] splines_3.3.2
## [45] GenomicRanges_1.26.3
## [46] SummarizedExperiment_1.4.0
## [47] quadprog_1.5-5
## [48] stringi_1.1.2
## [49] survey_3.31-5
## [50] RCurl_1.95-4.8
```